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Research Article

Achieving Compact and Uniform Buried Contact via Sequentially Deposited Hybrid Self-Assembled Monolayers for Efficient Wide-Bandgap Perovskite and All-Perovskite Tandem Solar Cells

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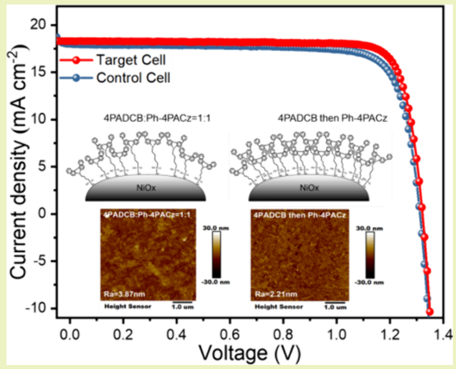
ABSTRACT: Wide-bandgap (WBG) perovskites have emerged as promising materials for all-perovskite tandem solar cells (ATSCs) for their potential to surpass the Shockley−Queisser limit of single-junction perovskite solar cells (PSCs). However, nonradiative recombination at the buried interface of WBG PSCs remains a great challenge, limiting the efficient carrier transport and collection in these devices. Hole selecting materials (HSMs) play a crucial role in charge extraction and growth of the overlying perovskite films. Here, we systematically investigated the deposition of various self-assembled monolayer (SAM) materials on NiO_x to modify the NiO_x/perovskite interface. A sequential deposition of (4-(7H-dibenzo[c,g]-carbazol-7-yl)butyl)phosphonic acid (4PADCB) and [4-(3,6-diphenyl-9H-carbazol-9-yl)butyl]-phosphonic acid (Ph-4PACz) is found to result in optimized efficiency, which can be attributed to the ability of sequential deposition to fill the gaps and form a more compact and uniform buried interfacial contact compared to coating each layer separately. Moreover, the modified buried interface notably eliminates the formation of the small grains, which may be caused by random nucleation. This strategy also enhances energy level alignment, decreasing the barrier for carrier transport at the buried interface. As a result, the champion single-junction WBG PSC achieves an open-circuit voltage (V_{OC}) of 1.33 V and an efficiency of 20.35%. The ATSCs fabricated with the WBG subcells based on the reported strategy achieve an optimized efficiency of 27.03%, exhibiting the great potential of the sequential deposition method for SAMs on future perovskite photovoltaics.



KEYWORDS: *sequential deposition, self-assembled monolayers, tandem perovskite solar cells, buried interface, wide-bandgap*

1. INTRODUCTION

Perovskite solar cells (PSCs) have drawn extensive attention over the past decade due to their exceptional optoelectronic properties,[1] with single-junction PSCs achieving a power conversion efficiency (PCE) of 26.8%.[2] To further surpass the efficiency limit of single-junction cells, perovskite tandem solar cells have been developed rapidly in recent years, among which all-perovskite tandem solar cells (ATSCs) show great potential due to their tunable bandgap range (1.1−2.3 eV).[3] Currently, ATSCs using wide-bandgap (WBG) perovskite materials as the top subcell have achieved a record efficiency exceeding 28%.[4] However, despite their promising prospects, ATSCs face many challenges, particularly the nonradiative recombination at buried interfaces, which remains a critical bottleneck limiting performance enhancement.[5] The high defect density, energy level mismatches, and interfacial reactions at buried interfaces exacerbate charge recombination, resulting in open-circuit voltage (V_{OC}) losses.[6] Enhancing the buried interface to improve carrier transport and interfacial performance is essential for enhancing ATSC performance.[7]

Self-assembled monolayers (SAMs) have demonstrated great potential as an effective interface modification technique for reducing interface defects, improving energy level alignment, and enhancing charge extraction efficiency.[8,9] In recent years, much research has focused on improving the interface between perovskite and hole selecting materials (HSMs), with nickel oxide (NiO_x) widely used as an HSM in WBG PSCs.[10] Despite their advantages of low-temperature fabrication, excellent chemical stability, and high optical transmittance, defects at the NiO_x/perovskite interface result in low charge extraction efficiency and significant V_{OC} losses. To address these issues, researchers have attempted to deposit SAMs onto NiO_x to improve the interface between NiO_x and the

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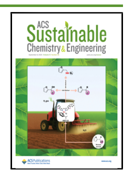

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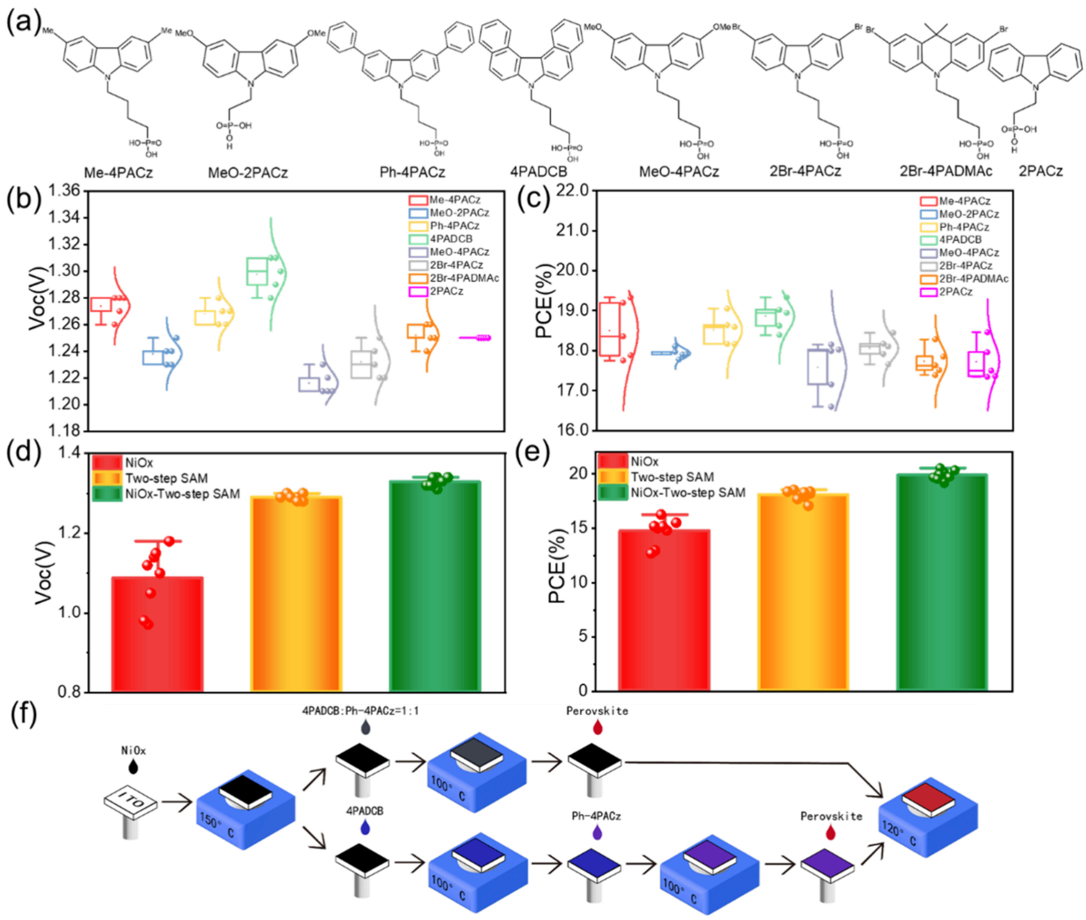

Figure 1. SAM material selection and sequential deposition experimental process. (a) Self-assembled monolayer (SAM) layer structure of different materials. (b) Open-circuit voltage (V_{OC}) values for perovskite solar cells with various SAM layers. (c) Power conversion efficiency (PCE) for the same cells indicating performance variation with different SAM treatments. (d) Comparison of V_{OC} for cells with only NiO_x, sequential deposition, and NiO_x combined with sequential deposition. (e) PCE comparison under the same conditions. (f) Schematic of the fabrication process for perovskite solar cells using NiO_x and different SAM layers, detailing the steps and temperatures involved.

perovskite layer. For instance, carbazole-based SAMs such as [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz) and [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) have been widely applied in NiO_x/perovskite heterojunctions due to their excellent molecular design and tunability.[11] Another example involves carbazole-based SAMs such as [4-(3,6-diphenyl-9H-carbazol-9-yl)butyl]phosphonic acid (Ph-4PACz), which, by extending the $\pi-\pi$ conjugation range through connected benzene rings, reduces defect density on the NiO_x surface and improves the perovskite crystallization process, substantially enhancing V_{OC} and PCE. These advancements provide new solutions for achieving high-performance and stable large-area perovskite devices.[12,13] Although these methods significantly improve the fill factor (FF) and PCE, they also have limitations. Carbazole-based SAMs, due to their amphiphilic nature, tend to aggregate in solvents, hindering the formation of a monolayer on NiO_x and leaving numerous voids, thereby failing to effectively passivate the deep defects in perovskites.[14]

In this study, we adopted a sequential deposition method to enhance the buried interface contact in WBG PSCs by successively depositing (4-(7H-dibenzo[c,g]-carbazol-7-yl)-butyl)phosphonic acid (4PADCB) and Ph-4PACz onto NiO_x. Compared with single-layer SAMs, this sequential deposition strategy addresses the issue of molecular aggregation observed in carbazole-based SAMs by forming a tightly packed molecular arrangement, effectively filling interfacial voids and ensuring better monolayer formation. Furthermore, this method significantly improves the coverage of the interface, resulting in the more efficient passivation of defects and optimized charge transport. Meanwhile, previous studies have explored bilayer SAM strategies, such as sequentially applying 2PACz and Me-4PACz.[15] Recently, Li et al. reported a coadsorbed SAM strategy using 2PACz and a small molecule additive to reduce aggregation and improve interface smoothness and work function in p−i−n PSCs and OSCs.[16] However, this coadsorption limits precise control of molecular packing and uniformity. In contrast, our work targets NiO_x surfaces in wide-bandgap and tandem PSCs, systematically comparing SAM materials, mixing methods, and deposition sequences. We found that depositing 4PADCB first and then Ph-4PACz forms a more compact, uniform buried interface with lower surface roughness and better energy alignment, leading to superior device performance. Experimentally, this method enabled single-junction WBG PSCs to achieve a V_{OC} of 1.33 V and a PCE of 20.35% with a bandgap of 1.77 eV. Moreover, ATSCs based on this approach achieved an impressive efficiency of 27.03%. These results show that the sequential deposition method has great potential in the field of perovskite photovoltaics, especially in improving

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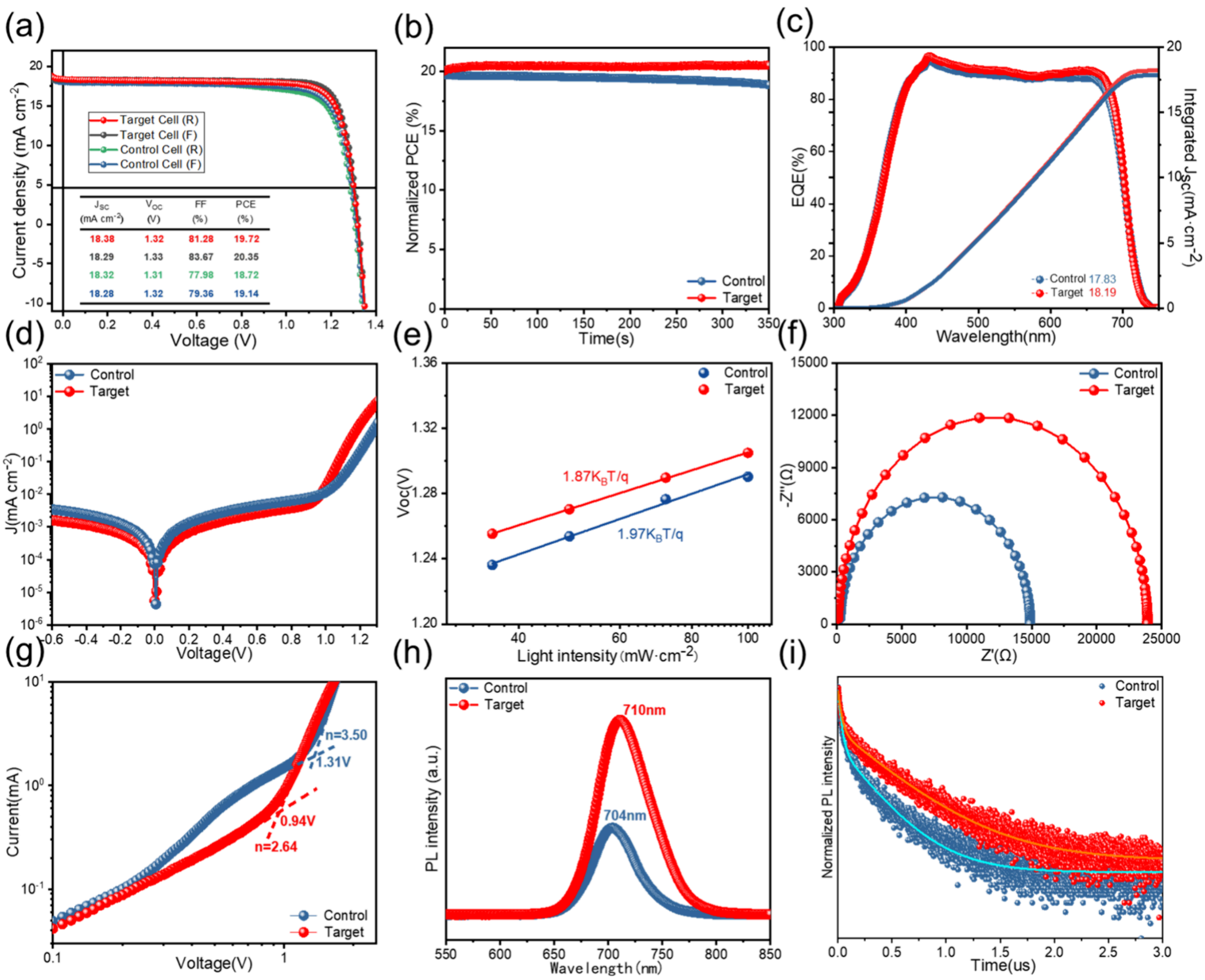

Figure 2. Optical and electronic characterization of PSCs modified by the 1:1 mixed method and sequential deposition method. (a) Current density−voltage (J−V) curves of the best-performing devices in the control and target groups. (b) Power output tracking at the maximum power point (MPP) over time for both control and target devices. (c) External quantum efficiency (EQE) curves for the control and target devices at different wavelengths. (d) Dark J−V curves for both groups. (e) The light intensity-dependent J−V curves. (f) Nyquist plots showing differences in impedance characteristics. (g) Space-charge-limited current (SCLC) measurements displaying the current−voltage characteristics under different electric field strengths. (h) Steady-state photoluminescence (PL) spectra indicating peak emission wavelengths. (i) Time-resolved photoluminescence (TRPL) decay profiles demonstrating the lifetime of photogenerated carriers.

interface properties and reducing nonradiative recombination. Future research can explore further improving device efficiency by optimizing the design of SAM molecules and deposition process and achieving large-area production and long-term stability. In addition, the scalability and commercialization potential of this method are also worthy of further verification, providing new solutions for the practical application of perovskite photovoltaic technology.

2. RESULT AND DISCUSSION

2.1. Sequential Deposition of Two Self-Assembled Monolayer Layers. We systematically investigated the impact of different self-assembled monolayer (SAM) treatments on the performance of single-junction wide-bandgap (WBG) perovskite solar cells (PSCs). By evaluating the open-circuit voltage (V_{OC}) and high power conversion efficiency (PCE) across various SAM materials and configurations, we aimed to determine the best combination for enhancing the device performance. We used a range of different materials for the SAM layer (Figure 1a), including [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz), [2-(3,6-dimethoxy-9H-carbazole-9-yl)ethyl]phosphonic acid (MeO-2PACz), [4-(3,6-diphenyl-9H-carbazol-9-yl)butyl]phosphonic acid (Ph-4PACz), (4-(7H-dibenzo[c,g]-carbazol-7-yl)butyl)-phosphonic acid (4PADCB), [4-(3,6-dimethoxy-9H-carbazol-9-yl)butyl]phosphonic acid (MeO-4PACz), 2-bromo-4-phosphonobutanoic acid (2Br-4PACz), 2-bromo-4-phosphonobutanoic acid (2Br-4PADMAc), and [2-(9H-carbazol-9-yl)ethyl]-phosphonic acid (2PACz). Figure 1b shows the V_{OC} values for PSCs employing a range of SAM layers. The results demonstrate significant variability in V_{OC} depending on the SAM material used. Notably, the cells treated with 4PADCB exhibited the highest average V_{OC} among all samples,

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indicating superior charge extraction and reduced recombination losses, followed by those of Me-4PACz and Ph-4PACz.

As shown in Figure 1c, the PCEs of the same cells mirrored the trends observed in the V_{OC}. The cells incorporating 4PADCB, Me-4PACz, and Ph-4PACz achieved the highest PCE values, reflecting enhanced device performance due to improved interface quality and refined charge carrier dynamics.

To further improve device performance, we investigated the effects of mixing different SAM materials in pairs (Figure S1). The results obtained from combining 4PADCB with Ph-4PACz, 4PADCB with Me-4PACz, and Ph-4PACz with Me-4PACz in a 1:1 ratio showed that the combination of 4PADCB and Ph-4PACz performed the best, with relatively higher V_{OC}, fill factor (FF), and PCE values. Building on our previous research, we further explored the impact of sequentially depositing two different SAM layers on optimizing device performance (Figure S2), where one SAM material was applied first, followed by another. By presenting the results of different sequential combinations, we found that the best results were obtained by using 4PADCB first and then Ph-4PACz.

To further improve the performance of the sequential SAM layer strategy, we investigated the impact of different concentrations of 4PADCB and Ph-4PACz on the device efficiency (Figure S3 and Figure S4). We explored various concentrations of 4PADCB ranging from 1 mg/mL to 0.1 mg/mL while keeping the concentration of Ph-4PACz constant. The results showed that a 4PADCB concentration of 0.75 mg/mL yielded the best overall performance with the highest V_{OC}, FF, and PCE values. This concentration appears to provide the optimal balance between effective surface coverage and charge transport efficiency.[17,18] Additionally, we focused on the concentration variation of Ph-4PACz, ranging from 1.5 to 0.1 mg/mL. The data revealed that a concentration of 0.75 mg/mL for Ph-4PACz also resulted in the best device performance. This combination of 0.75 mg/mL for both 4PADCB and Ph-4PACz not only maximized V_{OC} and FF but also significantly enhanced the overall PCE.

We further analyzed the performance improvement resulting from the sequential deposition (applying 4PADCB first, followed by Ph-4PACz) compared to that of the traditional NiO_x layer and NiO_x combined with the sequential deposition. As shown in Figure 1d,e, using sequential deposition, whether alone or in combination with NiO_x, resulted in a significant increase in V_{OC} and PCE.

Figure 1f shows the process flow for sequential deposition. Contact angle measurements (Figure S5) confirmed that Ph-4PACz was effectively deposited on the 4PADCB layer. The contact angle on the NiO_x surface was 24.1°, while a 1:1 mixture of 4PADCB and Ph-4PACz on NiO_x showed 90.0°. When 4PADCB was deposited first, followed by Ph-4PACz, the contact angle increased to 96.6°, confirming the successful deposition of both layers.

2.2. Characterization of Wide-Bandgap Perovskite Films and Devices. To clarify the subsequent discussion, we refer to perovskite films or devices with the 1:1 4PADCB and Ph-4PACz mixed solution as the SAM layer as the control group, while those treated with the sequential deposition of applying 4PADCB first, followed by Ph-4PACz, are referred to as the target group. We developed single-junction WBG PSCs, featuring a device structure of ITO/NiO_x/hole transport layer/perovskite/Poly(1,3,5-triazine-2,4-diamine)-bromide (PiPBr)/lithium fluoride (LiF)/buckminsterfullerene (C60)/bathocuproine (BCP)/Cu (Figure S6). As shown in Figure 2a, the control cell achieved the best PCE of 19.14% (18.72%) under the forward (reverse) scan, with a V_{OC} of 1.32 V (1.31 V), a short-circuit current density (J_{sc}) of 18.28 mA cm^{-2} (18.32 mA cm^{-2}), and a FF of 79.36% (77.98%). In contrast, the target cell exhibited a best PCE of 20.35% (19.72%) under a forward (reverse) scan, with a V_{OC} of 1.33 V (1.32 V), a J_{sc} of 18.29 mA cm^{-2} (18.38 mA cm^{-2}), and an FF of 83.67% (81.28%).

To evaluate the long-term stability of both control and target devices, we performed maximum power point tracking (MPPT) measurements, as shown in Figure 2b. The data revealed a significant contrast in the stability of the two devices over an extended period. Notably, the target device exhibited a remarkable improvement in performance, with its PCE increasing to approximately 103% of its initial value after 350 s of continuous operation. In contrast, the control device decreased to 94%. This enhancement can be attributed to the sequential deposition (4PADCB followed by Ph-4PACz), which creates a stable and uniform interface, reducing nonradiative recombination and improving the crystallinity and structural integrity of the perovskite layer.[19−21]

Figure 2c shows the external quantum efficiency (EQE) spectra. The EQE curve of the target device outperforms that of the control group across the entire wavelength range, reaching a peak of 96% at 430 nm. The integrated current density of the target group is also higher than that of the control group.

Figure 2d shows the dark current density−voltage ($J-V$) characteristics of the control and target devices. In the low-voltage region on the left, the dark current density of the target device is slightly lower than that of the control device, indicating a reduced charge carrier recombination at low voltages.

To examine the effects of sequential deposition on charge recombination, we performed light intensity-dependent $J-V$ measurements. As illustrated in Figure 2e, V_{OC} shows a logarithmic increase as light intensity rises. This relationship was fitted using the following eq 1:

$$V_{OC} = \frac{nkT\ln(L)}{q} \tag{1}$$

In this equation, n, k, T, L, and q denote the ideality factor, Boltzmann constant, absolute temperature, light intensity, and elementary charge, respectively. When compared with the control device, the n value for the target device decreases from 1.97 to 1.87. This improvement contributes to the overall performance of the device, allowing the target device to demonstrate better voltage response at higher light intensities.[22]

We performed electrochemical impedance spectroscopy (EIS) measurements to gain further insight into the transport mechanisms of the WBG devices (Table S1). The Nyquist plot (Figure 2f) was recorded under dark conditions using a bias voltage of 1.18 V and a frequency sweep from 10 MHz to 10 Hz. The sequential deposition reduced the series resistance (R_s) of the target device from 262.1 to 100.9 Ω, while the recombination resistance (R_{rec}) increased from 14617 to 23806 Ω.

Figure 2g displays the space-charge-limited current (SCLC) characteristics of the control and target devices. The devices were fabricated with a structure of ITO/NiO_x/perovskite/poly(triarylamine) (PTAA)/Ag. The trap-filled limit voltage (V_{TFL}) for the control device is approximately 1.31 V, while for

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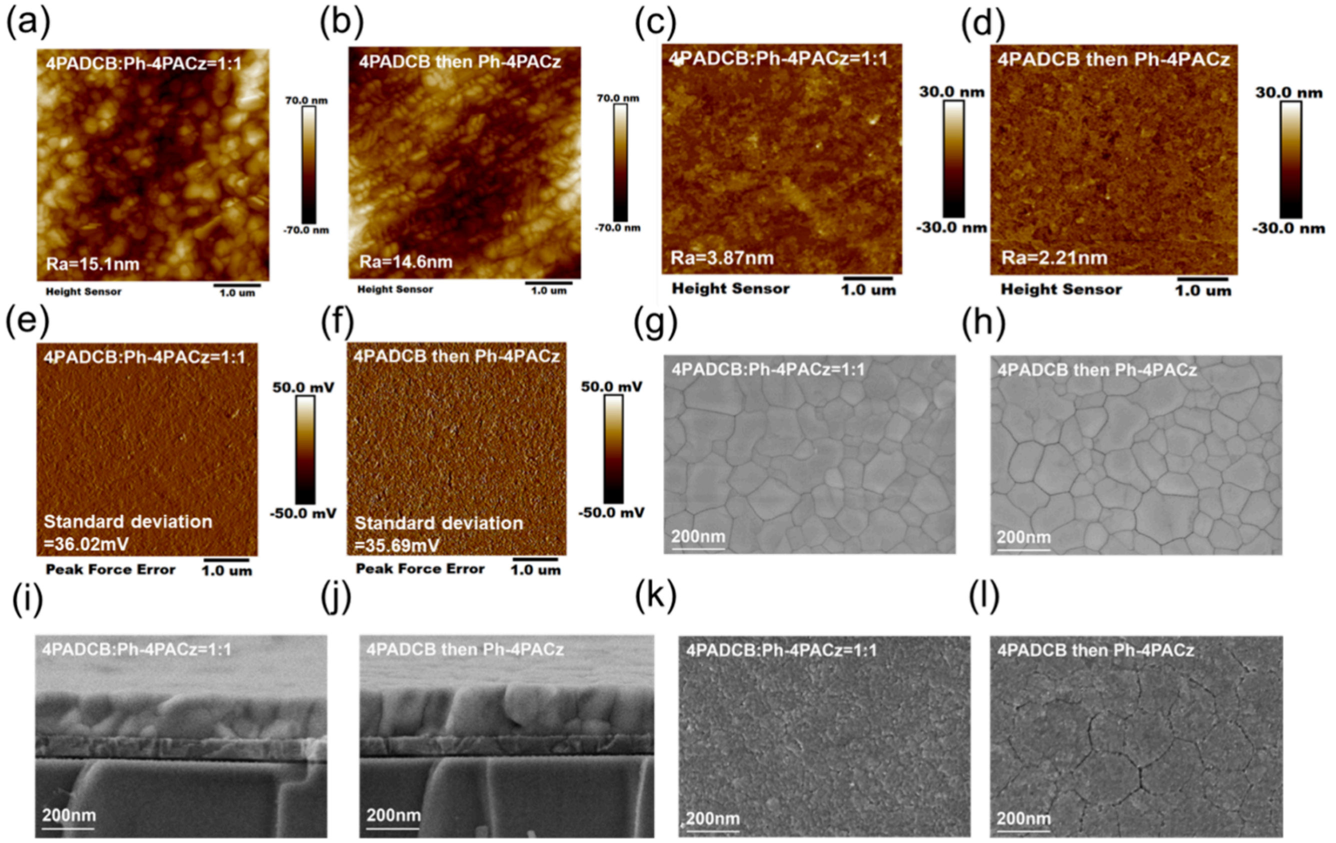

Figure 3. Structure characterization of the perovskite films. (a) Atomic force microscope (AFM) image of the control sample (4PADCB:Ph-4PACz = 1:1 SAM layer). (b) AFM image of the target sample (4PADCB and Ph-4PACz SAM layers). (c) AFM image of the perovskite buried interface of the control sample. (d) AFM image of the perovskite buried interface of the target sample. (e) Kelvin probe force microscopy (KPFM) image of the control sample. (f) KPFM image of the target sample. (g) Top-view scanning electron microscope (SEM) image of the control sample. (h) Top-view SEM image of the target sample. (i) Cross-section SEM image of the control sample. (j) Cross-section SEM image of the target sample. (k) Top-view SEM image of the perovskite buried interface of the control sample. (l) Top-view SEM image of the perovskite buried interface of the target sample.

the target device, it is significantly lower at 0.94 V. This reduction in V_{TFL} for the target device indicates a lower trap density in the perovskite layer.[23]

We also conducted steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) measurements to investigate the charge dynamics of the perovskite films. As shown in Figure 2h, the PL peak of the target sample is red-shifted from 704 to 710 nm, and its PL intensity is significantly higher than that of the control sample. This red shift is primarily attributed to a reduction in defect density and enhanced crystallinity, which leads to more efficient radiative recombination and reduced exciton trapping. Further TRPL analysis (Figure 2i and Table S2) shows that the average carrier lifetime of the target film is 501 ns, considerably higher than the 298 ns observed in the control sample. Halide segregation may lead to a shift or increase in the PL peak,[24] so we performed PL and TRPL measurements at different illumination times in Figure S7. These data show that no obvious phase separation or halide segregation occurs in the treated samples, supporting the observed PL changes and confirming that the red shift is basically unrelated to halide segregation. To further examine the interfacial charge dynamics, transient absorption (TA) spectroscopy was performed (Figure S8). The target sample exhibits a deeper and more confined photoinduced bleaching (PB) signal, indicating higher carrier density, more efficient charge extraction, and reduced nonradiative recombination at the buried interface.[25] These results collectively suggest that the target sample exhibits superior optoelectronic properties, with higher crystallinity and more effective suppression of nonradiative recombination.

2.3. Perovskite Film Growth and Morphology Optimization. The atomic force microscopy (AFM) images show the effects of different SAM layer deposition methods on the surface morphology of perovskite films. In Figure 3a, the film treated with a 1:1 mixed solution of 4PADCB and Ph-4PACz has a surface roughness (Ra) of 15.1 nm, while the film in Figure 3b, processed using the sequential deposition, has a slightly smoother surface with a Ra of 14.6 nm. Figure S9a presents the AFM image of the treated NiO_x substrate, further illustrating the effects of these treatments.

We also measured the buried interface of the perovskite layer (Figure 3c,d), and the results show that the sequential deposition reduced the surface roughness from 3.87 nm in the control sample to 2.21 nm in the target sample.

We further present the Kelvin probe force microscopy (KPFM) images of the surface potential distribution of the perovskite films (Figure S9b, Figure 3e,f). The results show that the sample treated with the sequential deposition has a surface potential distribution with a standard deviation of 35.69 mV, which is lower than the 36.02 mV observed for the control sample. The KPFM image in Figure S10 also shows the

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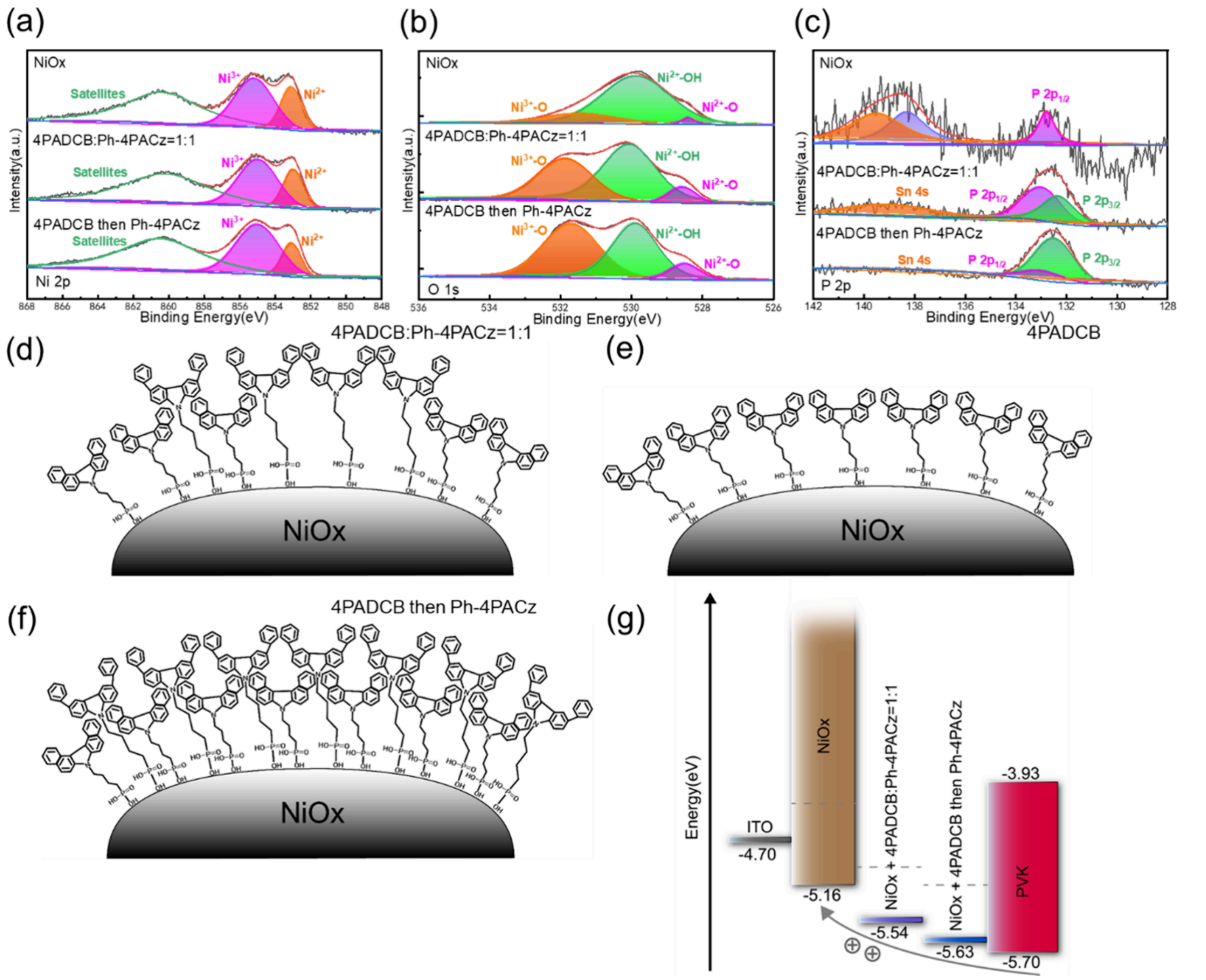

Figure 4. Mechanism study of the SAM layer under sequential deposition. XPS spectra of (a) Ni *2p*, (b) O *1s*, (c) and P *2p* core levels for NiO_x, control (4PADCB:Ph-4PACz = 1:1), and target (4PADCB then Ph-4PACz) samples. Schematic diagram of the sequential deposition interface engineering mechanism is shown in (d). Using a 1:1 mixed solution of 4PADCB and Ph-4PACz on NiO_x results in disordered molecular arrangement and interfacial voids. (e) A single layer of 4PADCB deposited on the NiO_x surface still results in uneven coverage, leaving gaps that hinder charge transport. (f) Using sequential deposition, 4PADCB is deposited first and then Ph-4PACz, forming a compact and ordered molecular arrangement, reducing voids and improving interface uniformity. (g) Energy level arrangement diagram of NiO_x and perovskite layers under different interface treatment methods. The dashed line indicates the Fermi level.

potential distribution of the buried interface, with the standard deviation decreasing from 35.60 to 35.08 mV.

Top-view scanning electron microscopy (SEM) images (Figure S11a, Figure 3g,h) show no significant changes in the surface morphology between the films treated with the sequential deposition method and the control sample. The cross-sectional SEM images (Figure S11b, Figure 3i,j) also exhibit no noticeable differences in the crystal structure. However, the buried interface (Figure 3k,l) reveals significant changes, with the random nucleation-induced small grains being effectively eliminated, resulting in much larger grains compared to the control sample. This indicates that sequential deposition notably improves the crystallinity and uniformity of the perovskite layer at the buried interface.

To further confirm the changes in film crystallinity caused by different SAM treatments, we also performed X-ray diffraction (XRD) measurements (Figure S12), and the diffraction patterns showed enhanced crystallinity after SAM treatment with clearer and sharper peaks compared to bare NiO_x. Notably, the sequentially deposited samples exhibited distinct diffraction features, indicating improved molecular ordering and better structural compatibility with subsequent perovskite growth. The absorption spectra in Figure S13show little difference between the NiO_x and SAM-treated substrates, indicating that these treatments have a minimal impact on the light absorption properties.

The chemical states of NiO_x thin films under different treatment conditions were investigated by using X-ray photoelectron spectroscopy (XPS). For NiO_x films, both in the control group with a 1:1 mixed SAM layer and in the target group with sequential deposition, the Ni *2p* spectra showed peaks at 855.1 and 853.1 eV, corresponding to Ni^{3+} and Ni^{2+}, respectively (Figure 4a). Ni^{3+}, primarily located at grain boundaries in oxygen-rich, nonstoichiometric NiO_x, facilitates hole transport.[26] The Ni^{3+} proportion increased from 54% in the control group to 61% in the target group, indicating that the sequential deposition effectively enhances Ni^{3+} content and improves hole transport.[27]

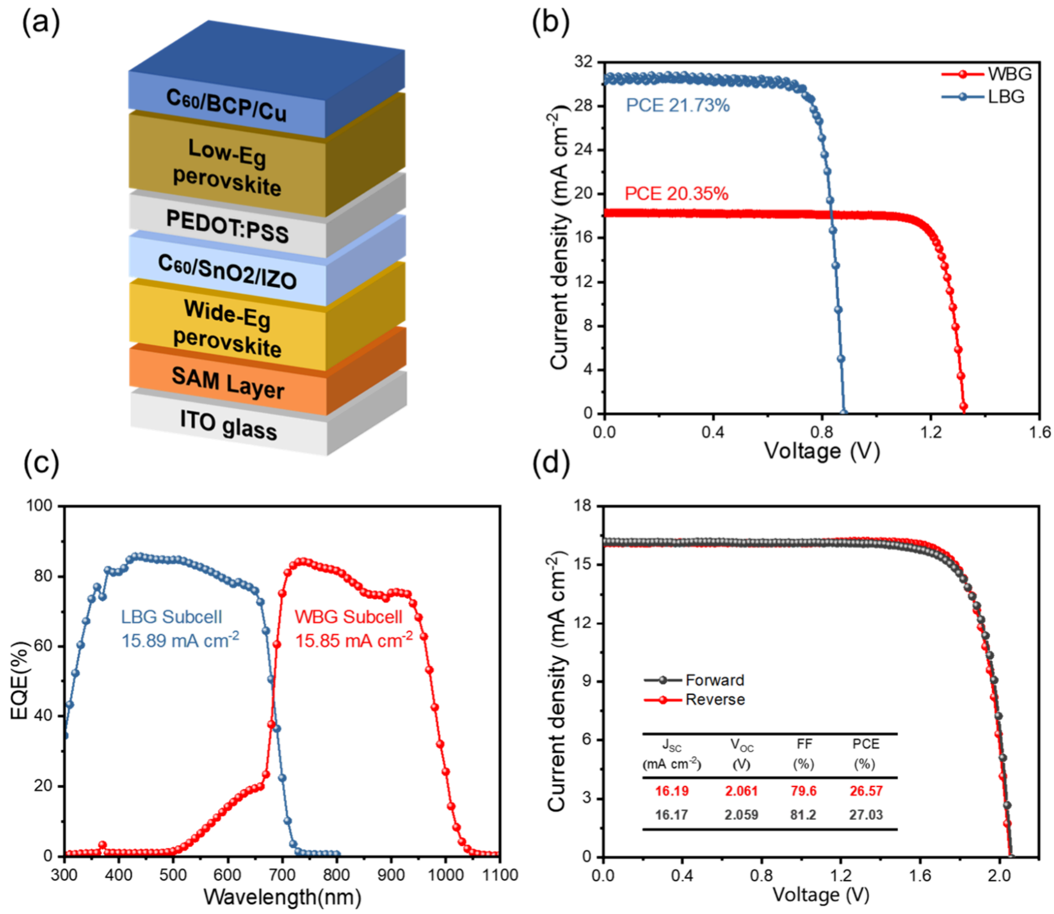

Figure 5. Photovoltaic performance and stability testing based on all-perovskite tandem solar cells. (a) The device architecture of perovskite tandem solar cells. (b) The J–V curves of the low-bandgap (LBG) and wide-bandgap (WBG) subcells with PCE values of 21.73% and 20.35%. (c) EQE curves of LBG and WBG subcells. (d) Current density–voltage (J–V) characteristics of the champion all-perovskite tandem solar cell measured under both forward and reverse scan directions. The device exhibits a J_{sc} of 16.17 (16.19) mA cm^{-2}, a V_{OC} of 2.059 (2.061) V, a FF of 81.2% (79.6%), and a PCE of 27.03% (26.57%) under forward (reverse) scan.

The O *1s* spectrum revealed peaks at 529.9, 531.7, and 528.5 eV, corresponding to Ni^{2+}–OH, Ni^{3+}–O, and Ni^{2+}–O, respectively (Figure 4b). In untreated samples, Ni^{2+}–OH accounted for 87%, while Ni^{3+}–O was only 8%. After 1:1 mixed SAM treatment, Ni^{2+}–OH decreased to 50% and Ni^{3+}–O increased to 37% (Table S3). With the sequential deposition, Ni^{2+}–OH further dropped to 43%, and Ni^{3+}–O rose to 44%, demonstrating that SAM treatment reduces Ni^{2+}–OH and increases Ni^{3+}–O, enhancing conductivity and device performance.[28] The P *2p* spectra showed peaks at 133 and 132.5 eV, corresponding to the $2p_{1/2}$ and $2p_{3/2}$ states, respectively (Figure 4c). The $2p_{3/2}$ proportion was 42% after 1:1 mixed SAM treatment, increasing to 79% with the sequential deposition, reflecting denser molecular packing and improved charge transport.[29] The Sn *3d* spectra (Figure S14) showed peaks for Sn^{4+} and Sn^{2+} at binding energies of 494.8 and 486.3 eV, respectively. The Sn^{4+} proportion increased from 42% in untreated samples to 45% with 1:1 mixed SAM treatment and 47% with sequential deposition, indicating improved interfacial chemical states and electrical properties.[30] At the buried interface, the P *2p* and O *1s* spectra (Figure S15) revealed a significantly higher $2p_{3/2}$ proportion and an increased Ni^{3+}–O content in samples treated with sequential deposition, confirming enhanced surface activity and electron transport efficiency.

Mechanism models (Figure 4d–f) showed that 1:1 mixed SAM treatment resulted in disordered molecular alignment and interfacial voids, while single-step 4PADCB treatment left uneven regions. The sequential deposition formed tightly packed Ph-4PACz molecules over the 4PADCB layer, reducing voids, improving potential distribution, and enhancing charge transport and device performance.[31]

Ultraviolet photoelectron spectroscopy (UPS) results (Figure S16) showed that the energy difference between the Fermi level and the valence band maximum (VBM) decreased from 0.49 eV in the control sample to 0.47 eV with the sequential deposition, facilitating hole transport, improving interfacial energy alignment, and reducing the energy mismatch.

A moderate improvement in the device stability was also observed. As shown in Figure S17, the PCE of the champion device maintained over 85% of its initial value after 900 h in ambient conditions. Energy level alignment studies (Figure 4g) indicated that the VBM difference between NiO_x and the perovskite layer reduced from 0.54 eV in untreated samples to 0.16 eV with 1:1 mixed SAM treatment and further to 0.07 eV with the sequential deposition, significantly improving interfacial charge transport.[32]

2.4. All-Perovskite Tandem Solar Cells Based on 4PADCB Interface Engineering. We fabricated monolithic all-perovskite tandem solar cells using 4PADCB interface

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engineering for the WBG subcell. The device architecture is shown in Figure 5a, consisting of low-band gap (LBG) and WBG perovskite layers. This tandem structure is designed to refine photovoltaic performance by enhancing cell efficiency through interface engineering.[33] The tandem devices were fabricated by sequentially preparing the WBG and LBG subcells, with the WBG perovskite layer processed as previously described. The interconnecting layer consisted of atomic layer deposition (ALD)-deposited SnO_2 and sputtered IZO. The LBG perovskite layer was prepared via spin coating, followed by $EDAI_2$ surface treatment and vacuum deposition of charge transport and electrode layers. Figure 5b shows the $J-V$ curves of the LBG and WBG subcells. The LBG single-junction cell achieved a PCE of 21.73%, while the WBG single-junction cell achieved a PCE of 20.35%. The integrated J_{sc} values derived from the EQE spectra for the WBG and LBG subcells are 15.85 and 15.89 mA cm^{-2}, respectively, aligning closely with the J_{sc} values obtained from $J-V$ measurements (Figure 5c). We performed cross-sectional SEM imaging to confirm the vertical stacking structure of the tandem device. As shown in Figure S18, the WBG and LBG perovskite layers are clearly distinguished, with sharp interfaces and uniform morphology, indicating good layer continuity and structural integrity. As shown in Figure 5d, the champion tandem solar cell exhibited a J_{sc} of 16.17 mA cm^{-2} (16.19 mA cm^{-2}), a V_{OC} of 2.059 V (2.061 V), an FF of 81.2% (79.6%), and a PCE of 27.03% (26.57%) under forward (reverse) scan. These parameters indicate that the tandem device achieves high efficiency while maintaining a good voltage and current output. The $J-V$ curves show little hysteresis between the forward and reverse scan, indicating that the charge transport and extraction processes of the device are well refined. The long-term stability of the device was further evaluated using MPPT, and as shown in Figure S19, the PCE remains stable over a 380-h period.

3. CONCLUSIONS

In conclusion, we have demonstrated significant performance enhancements in PSCs through the implementation of sequential deposition, where 4PADCB is first applied to the NiO_x surface, followed by Ph-4PACz. By optimizing this interface between the SAM layers and the NiO_x surface, we have achieved notable improvements in key parameters, including V_{OC}, FF, and PCE. The champion device fabricated by using this approach achieved an impressive PCE of 20.35%. These results underscore the effectiveness of the sequential deposition in enhancing the performance of PSCs. Furthermore, we successfully applied this strategy to fabricate tandem solar cells. The all-perovskite tandem devices achieved exceptional efficiencies, with the champion tandem device reaching a PCE of 27.03%. The integration of sequential deposition contributed significantly to the improved current matching and overall device performance, as evidenced by the increased V_{OC} of 2.059 V and stable power output over extended operation. These findings highlight the potential of this sequential deposition for advancing the development of efficient multijunction tandem solar cells, opening up new opportunities for further optimization and commercial applications in the field of perovskite photovoltaics.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.5c03613.

Experimental section including materials, device fabrication, characterization, TA, PL, TRPL, EQE, $J-V$ characteristics, contact angle, photovoltaic (PV) parameters, XPS, XRD, AFM, KPFM, UPS, and SEM (PDF)

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Author Contributions

Z.Y. conceived the idea. Z.Y., L.Z., and Z.H. designed the experiments. Z.H. fabricated all the devices and conducted the characterization. T.M. and C.W. fabricated and measured the tandem solar cells device. L.Z., S.W., Z.H., Z.Z., and S.L.

measured and analyzed the XPS, UPS, Abs., PL, and TRPL for the perovskites. L.Z., Z.H., and X.W. measured and analyzed the XRD. Z.H. and L.Z. measured and analyzed TA. Z.H. and S.W. measured and analyzed the contact angle. L.Z., S.W., Y.X., and Z.H. measured and analyzed the SEM and the EQE spectra. Z.Z. and S.L. conducted device performance tests. L.Z., S.D., and Z.H. measured and analyzed the AFM and KPFM. Z.Y., Z.H., L.Z., S.W., and C.W. wrote the paper. All authors reviewed the paper.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Manser, J. S.; Christians, J. A.; Kamat, P. V. Intriguing optoelectronic properties of metal halide perovskites[J]. *Chem. Rev.* **2016**, *116* (21), 12956−13008.

(2) Liu, L.; Zheng, D.; Du, M.; et al. Advancements and Challenges in Wide-Bandgap Perovskite Solar Cells: From Single Junction to Tandem Solar Cells. *Sol. RRL* **2024**, *8* (17), No. 2400359.

(3) Shi, J.; Li, Y.; Li, Y.; et al. From ultrafast to ultraslow: charge-carrier dynamics of perovskite solar cells[J]. *Joule* **2018**, *2* (5), 879−901.

(4) Wang, Y.; Lin, R.; Liu, C.; et al. Homogenized contact in all-perovskite tandems using tailored 2D perovskite. *Nature* **2024**, *635*, 867−873.

(5) Gao, Z. W.; Wang, Y.; Choy, W. C. H. Buried Interface Modification in Perovskite Solar Cells: A Materials Perspective. *Adv. Energy Mater.* **2022**, *12* (20), No. 2104030.

(6) Yang, X.; Luo, D.; Xiang, Y.; et al. Buried Interfaces in Halide Perovskite Photovoltaics. *Adv. Mater.* **2021**, *33* (7), No. 2006435.

(7) Yuan, S.; Liu, Y.; Lan, J.; et al. Accurate dimension prediction for low-dimensional organic−inorganic halide perovskites *via* a self-established machine learning strategy[J]. *J. Phys. Chem. Lett.* **2023**, *14* (32), 7323−7330.

(8) Zhang, X.; Ma, Q.; Wang, Y.; et al. Ligand Homogenized Br−I Wide-Bandgap Perovskites for Efficient NiO_x-Based Inverted Semi-transparent and Tandem Solar Cells[J]. *ACS Nano* **2024**, *18* (24), 15991−16001.

(9) Hang, P.; Kan, C.; Li, B.; et al. Highly Efficient and Stable Wide-Bandgap Perovskite Solar Cells via Strain Management. *Adv. Funct. Mater.* **2023**, *33* (11), No. 2214381.

(10) Li, B.; Zhang, C.; Gao, D.; et al. Suppressing Oxidation at Perovskite−NiO x Interface for Efficient and Stable Tin Perovskite Solar Cells. *Adv. Mater.* **2024**, *36* (17), No. 2309768.

(11) Hung, C. M.; Wu, C. C.; Yang, Y. H.; et al. Repairing Interfacial Defects in Self-Assembled Monolayers for High-Efficiency Perovskite Solar Cells and Organic Photovoltaics through the SAM@Pseudo-Planar Monolayer Strategy. *Adv. Sci.* **2024**, *11* (36), No. 2404725.

(12) Sun, A.; Tian, C.; Zhuang, R.; et al. High Open-Circuit Voltage (1.197 V) in Large-Area (1 cm 2) Inverted Perovskite Solar Cell via Interface Planarization and Highly Polar Self-Assembled Monolayer. *Adv. Energy Mater.* **2024**, *14* (8), No. 2303941.

(13) Wu, T.; Mariotti, S.; Ji, P.; et al. Self-Assembled Monolayer Hole-Selective Contact for Up-Scalable and Cost-Effective Inverted Perovskite Solar Cells. *Adv. Funct. Mater.* **2024**, *34*, No. 2316500.

(14) Kim, S. Y.; Cho, S. J.; Byeon, S. E.; et al. Self-Assembled Monolayers as Interface Engineering Nanomaterials in Perovskite Solar Cells. *Adv. Energy Mater.* **2020**, *10* (44), No. 2002606.

(15) Cui, H.; Huang, L.; Zhou, S.; et al. Lead halide coordination competition at buried interfaces for low V_{OC}-deficits in wide-bandgap perovskite solar cells[J]. *Energy Environ. Sci.* **2023**, *16* (12), 5992−6002.

(16) Li, D.; Lian, Q.; Du, T.; et al. Co-adsorbed self-assembled monolayer enables high-performance perovskite and organic solar cells. *Nat. Commun.* **2024**, *15* (1), 7605.

(17) Shin, S. S.; Lee, S. J.; Seok, S. I. Metal Oxide Charge Transport Layers for Efficient and Stable Perovskite Solar Cells. *Adv. Funct. Mater.* **2019**, *29* (47), No. 1900455.

(18) Zheng, J.; Wei, H.; Ying, Z.; et al. Balancing Charge-Carrier Transport and Recombination for Perovskite/TOPCon Tandem Solar Cells with Double-Textured Structures. *Adv. Energy Mater.* **2023**, *13* (5), No. 2203006.

(19) Tao, J.; Zhao, C.; Wang, Z.; et al. Suppressing non-radiative recombination for efficient and stable perovskite solar cells[J]. *Energy Environ. Sci.* **2025**, *18* (2), 509−544.

(20) Baishya, H.; Adhikari, R. D.; Patel, M. J.; et al. Defect mediated losses and degradation of perovskite solar cells: Origin, impacts and reliable characterization techniques. *J. Energy Chem.* **2024**, *253* (217), 2095−4956.

(21) Liu, C.; Chen, J.; Gao, P. Enhanced perovskite photovoltaics with non-conjugated polymers: recent developments and future prospects[J]. *ACS Applied Polymer Materials* **2024**, *6* (7), 3573−3610.

(22) Bisquert, J.; Gonzales, C.; Guerrero, A. Transient on/off photocurrent response of halide perovskite photodetectors[J]. *J. Phys. Chem. C* **2023**, *127* (43), 21338−21350.

(23) Liu, Y.; Xu, T.; Xu, Z.; et al. Defect Passivation and Lithium Ion Coordination Via Hole Transporting Layer Modification for High Performance Inorganic Perovskite Solar Cells. *Adv. Mater.* **2024**, *36* (4), No. 2306982.

(24) Datta, K.; van Gorkom, B. T.; Chen, Z.; et al. Effect of light-induced halide segregation on the performance of mixed-halide perovskite solar cells[J]. *ACS applied energy materials* **2021**, *4* (7), 6650−6658.

(25) Wang, C.; Chu, W.; Ye, F.; et al. Polar methylammonium organic cations detune state coupling and extend hot-carrier lifetime in lead halide perovskites[J]. *Chem.* **2022**, *8* (11), 3051−3063.

(26) Yu, S.; Xiong, Z.; Zhou, H.; et al. Homogenized NiO_x nanoparticles for improved hole transport in inverted perovskite solar cells[J]. *Science* **2023**, *382* (6677), 1399−1404.

(27) Cai, X.; Hu, T.; Hou, H.; et al. A review for nickel oxide hole transport layer and its application in halide perovskite solar cells[J]. *Materials Today Sustainability* **2023**, *23*, No. 100438.

(28) Wang, S.; Li, Y.; Yang, J.; et al. Critical Role of Removing Impurities in Nickel Oxide on High-Efficiency and Long-Term Stability of Inverted Perovskite Solar Cells. *Angew. Chem.* **2022**, *134* (18), No. e202116534.

(29) Xu, Y.; Fei, F.; Dong, X.; et al. Uniform Coverage Functional Layers Enable High-Efficient Flexible Perovskite Solar Modules with an Outstanding Fill Factor. *Sol. RRL* **2023**, *7* (16), No. 2300283.

(30) Chowdhury, T. H.; Reo, Y.; Yusoff, A. R. B. M.; et al. Sn-Based Perovskite Halides for Electronic Devices. *Adv. Sci.* **2022**, *9* (33), No. 2203749.

(31) Chen, H.; Maxwell, A.; Li, C.; et al. Regulating surface potential maximizes voltage in all-perovskite tandems[J]. *Nature* **2023**, *613* (7945), 676−681.

(32) Iqbal, Z.; Zu, F.; Musiienko, A.; et al. Interface modification for energy level alignment and charge extraction in CsPbI3 perovskite solar cells[J]. *ACS Energy Letters* **2023**, *8* (10), 4304−4314.

(33) Zhao, C.; Zhang, H.; Krishna, A.; et al. Interface Engineering for Highly Efficient and Stable Perovskite Solar Cells. *Adv. Opt. Mater.* **2024**, *12* (7), No. 2301949.